TUO ZHANG

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EDUCATION

New York University, New York, NY Sep. 2023 - May 2026

Bachelor of Arts, Double Major in Mathematics and Computer Science

- Cumulative Undergraduate GPA (BU + NYU): **3.5 / 4.0**
- *Selected Coursework:* Machine Learning, Artificial Intelligence, Computer Graphics, Robotics, Operating Systems, Object-Oriented Programming, Analysis, Linear Algebra, Differential Geometry, Software Engineering

Boston University, Boston, MA – Dean's List Sep. 2021 - May 2023

Coursework in Mathematics and Computer Science (transferred to NYU in May 2023)

RELEVANT EXPERIENCE

CoCoBun AI Storybook Application – Gemini, Flux LoRA, FastAPI, Docker, GCP Jan. 2026 - Present

Full-stack Engineer United States

- Architected an end-to-end multimodal pipeline that turns a short parent prompt into a complete children's storybook, including structured story text, page-level captions, and matching illustrations
- Integrated a Flux diffusion model fine-tuned with LoRA into ComfyUI workflows to generate style-consistent characters and backgrounds across pages, suitable for printed or digital picture books
- Exposed the whole pipeline as a Dockerized FastAPI microservice with clean REST endpoints for story and image generation, deployable on Google Cloud so that web and mobile clients can call it over HTTP

Qrobot Jun. 2025 - Aug. 2025

Algorithm Engineer/Programmer Dongguan, China

- Developed industrial computer vision systems for semiconductor chip inspection using **OpenCV** and **PyQt5**
- Implemented LED/scratch/character detection and a multi-threaded GUI integrated with PLC communication, **reducing missed defects by ~55% on hold-out samples** and significantly improving inspection efficiency

PROJECTS

Machine Learning & Probabilistic Reasoning Projects – PyTorch, Python

- Implemented regression and classification models (logistic regression, SVMs, random forests) in **PyTorch**, achieving **~84% validation accuracy** on a high-dimensional dataset through regularization and systematic hyperparameter tuning
- Extended classical search algorithms (iterative deepening, hill climbing) to probabilistic reasoning and planning with **Bayesian networks, MDPs, and basic reinforcement learning**, analyzing decision-making behavior under noisy inputs

Containerized ML Monitoring System – Python, Flask, MongoDB, Docker, GitHub Actions

- Engineered a multi-container application composed of a machine learning client, a Flask web server, and a MongoDB database to support end-to-end ingestion, analysis, and visualization of machine-generated results
- Designed and integrated service communication across independently containerized subsystems using Docker Compose, improving modularity, reproducibility, and ease of local deployment
- Built a Flask + MongoDB web interface to display machine learning outputs, support persistent result storage, and enable structured retrieval of historical analysis records
- Established a basic CI/CD workflow with pytest and GitHub Actions to automate testing and improve development reliability before deployment

Autonomous Rover – Viam SDK, Python

- Fused RGB camera and LiDAR into a real-time perception and control stack on an autonomous rover, implementing color-based object following, obstacle avoidance, and trajectory recovery
- Tuned control parameters and safety logic to achieve **~99% successful completion** on predefined indoor routes with dynamic obstacles

Physically Based Rendering – Blender

- Built stylized 3D scenes in **Blender** and implemented physically based rendering algorithms, including ray tracing, path tracing, ambient occlusion, and Fresnel reflection, in Python to accurately simulate light–material interactions

SKILLS

- **Programming & Tools:** Python, C++, Java, C, Assembly, PyTorch, OpenCV, PyQt5, Viam SDK, Git, LaTeX
- **Languages:** Mandarin (Native), Cantonese (Fluent), English (Fluent)